NBTS
CODE-A

Aakash + BYJU'S

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MM : 720

NCERT Booster Program

Time : 3 hrs. 20 min

(*for* NEET-2023)

NBTS – 02

Topics covered :

Physics	:	Nuclei, Semiconductor, Laws of Motion, Work, Energy & Power
Chemistry	:	Biomolecules, Polymers, Chemistry in Everyday Life, Chemical Bonding and Molecular Structure, Hydrogen
Botany	:	Environmental Issues, The Living World, Biological Classification
Zoology	:	Biotechnology and Its Applications, Digestion and Absorption, Breathing and Exchange of Gases

Instructions :

(i) There are two sections in each subject, i.e. Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions from Section-B out of 15.

(ii) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered / unattempted questions will be given no marks.

(iii) Use blue/black ballpoint pen only to darken the appropriate circle.

(iv) Mark should be dark and completely fill the circle.

(v) Dark only one circle for each entry.

(vi) Dark the circle in the space provided only.

(vii) Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on the Answer sheet.

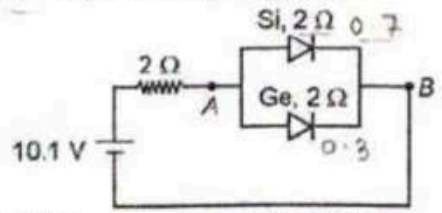
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PHYSICS

Choose the correct answer :

SECTION - A

1. If Si and Ge have cut in voltages as 0.7 V and 0.3 V respectively, then the voltage across points *A* and *B*, as shown in the figure, is



(1) 3.2 V (2) 0.3 V
(3) 6.4 V (4) 3.7 V

2. Which among the following relations is/are incorrect? (Symbols have their usual meaning)

(1) $\frac{1}{\alpha}-\frac{1}{\beta}=1$ (2) $\alpha=\frac{\beta}{\beta+1}$

(3) $\beta=\gamma-1$ (4) $\beta=\frac{\alpha}{\alpha+1}$

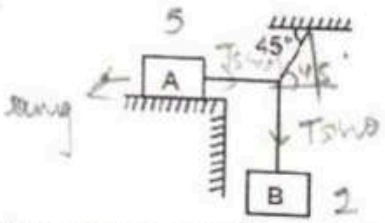
3. For a transistor amplifier, power gain and voltage gain are 100 and 5 respectively. The value of current gain for this amplifier will be

(1) 10 (2) 20
(3) 500 (4) 105

4. A block of mass 20 kg is placed on a long trolley. The coefficient of friction between the block and trolley is 0.2. The trolley is accelerating with acceleration 3 m/s². The acceleration of block with respect to trolley is (g = 10 m/s²)

(1) 5 m/s²
(2) 1 m/s²
(3) 2 m/s²
(4) 3 m/s²

5. Two blocks A and B of masses 5 kg and 1 kg are connected with the help of massless strings as shown in figure.



The system is just at the verge of slipping. The coefficient of static friction between the block A and surface below it is

(1) 0.2
(2) 0.15
(3) 0.18
(4) 0.25

6. Minimum horizontal force required to keep a block of mass 1 kg at rest against a rough vertical wall is N. Coefficient of friction between wall and block is 0.4. If a horizontal force $N/2$ is applied then acceleration of the block is

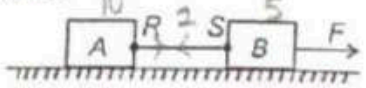
(1) 2 m/s²
(2) 3 m/s²
(3) 5 m/s²
(4) Zero

7. A circular racetrack of radius 200 m is banked at an angle 45°. If the angle of repose for wheels of car and racetrack is 30°, minimum permissible speed of car to avoid slipping is nearly

(tan 15° = 0.268, g = 10 m/s²)

(1) 86.39 m/s
(2) 44.25 m/s
(3) 33.15 m/s
(4) 23.15 m/s

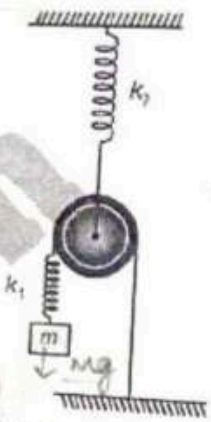
8. Two blocks A and B are kept on smooth horizontal surface. Both blocks are connected with a string as shown in figure. Mass of block A and B and string are 10 kg, 5 kg and 1 kg respectively.



If horizontal force F = 32 N is applied, then tension in string at point R will be

(1) 30 N
(2) 21.33 N
(3) 20 N
(4) 40 N

9. A block of mass m is connected with a spring (spring constant k_1) as shown in figure. Pulley is frictionless and massless.



When system is in equilibrium, then elongation in second spring (spring constant k_2) will be

(1) $\frac{mg}{k_1}$
(2) $\frac{2mg}{k_2}$
(3) $\frac{2mg}{k_1}$
(4) $\frac{mg}{k_2}$

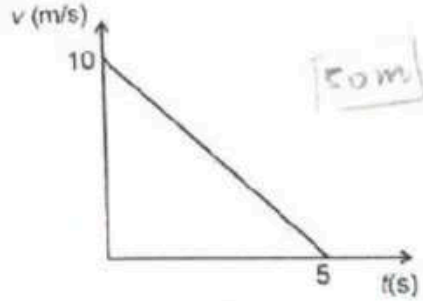
10. A stone of mass m tied to the end of a string revolves in a vertical circle of radius R. Tension in string at lowest point and highest point of the circle is T_1 and T_2. Minimum value of $T_1 + T_2$ is

(1) $6mg$
(2) $7mg$
(3) $8mg$
(4) $5mg$

11. Two billiard balls each of mass 50 g moving in opposite directions with speed 5 m/s collide and rebound with same speed. During collision, both balls are in contact for 0.05 s. Average force on each ball during collision is

(1) 20 N
(2) 100 N
(3) 50 N
(4) 10 N

12. A block of mass 1 kg is given a velocity 10 m/s horizontally on a rough horizontal plane. The following graph shows the velocity (v)-time (t) graph of the motion.



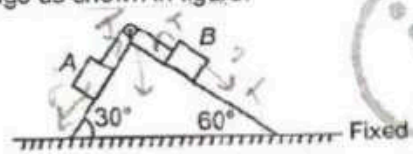
The co-efficient of kinetic friction between the plane and block is

($g = 10\ m/s^2$)

(1) 0.2 (2) 0.3

(3) 0.35 (4) 0.5

13. Two blocks A and B are in equilibrium on smooth wedge as shown in figure.



If mass of block A is 10 kg, then mass of block B is

(1) 30 kg (2) $10\sqrt{3}$ kg

(3) $\frac{10}{\sqrt{3}}$ kg (4) $\frac{10}{3}$ kg

14. A constant force $\vec{F} = (6\hat{i} + 2\hat{j})$ N acts on a particle of mass 2 kg. Particle is displaced from position $\vec{r}_1 = (2\hat{i} + \hat{j} + \hat{k})$ m to position $\vec{r}_2 = (3\hat{i} + 2\hat{j} + \hat{k})$ m.

The work done by the force on the particle is

(1) 2 J

(2) –2 J

(3) 4 J

(4) 8 J

15. Select the correct statement for semiconductor

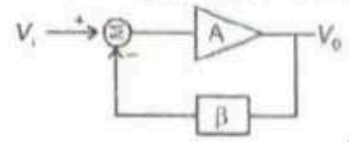
(1) It has infinite resistance at 0°C

(2) It has negative temperature coefficient of resistance

(3) Its band gap energy E_g is greater than 5 eV

(4) Its conductivity is greater than metals

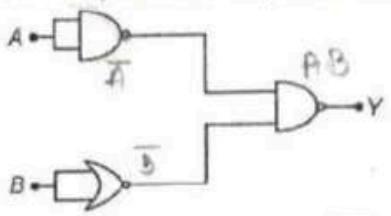
16. For the amplifier with feedback as shown in the figure, if $\beta = 0.05$ and overall gain (close loop gain) is 10, then open loop gain (A) of the amplifier is



(1) 100 (2) 20

(3) 40 (4) 5

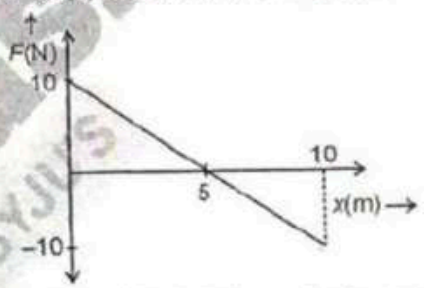
17. The combination of logic gates shown in the figure has output Boolean expression as



(1) $Y = A + B$ (2) $Y = \overline{A \cdot B}$

(3) $Y = A \cdot B$ (4) $Y = \overline{A + B}$

18. A force (F) acting on an object varies with distance (x) is shown in following figure

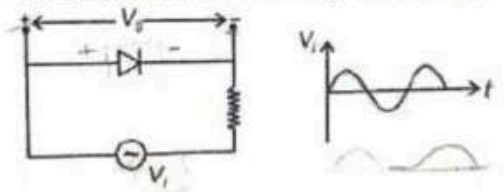


Mass of particle is 1 kg and at $x = 0$ kinetic energy of object is 5 J. At $x = 10$ m, kinetic energy of object is

(1) –5 J (2) 55 J

(3) 5 J (4) Zero

19. For the circuit shown in the figure, the output is



(1) Zero

(2) Half wave rectified with positive cycle in output

(3) Half wave rectified with negative cycle in output

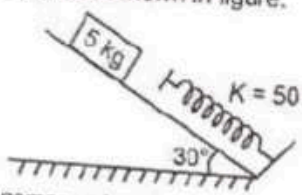
(4) Full wave rectified

20. Body A of mass $2m$ moving with speed u collides with another body B of mass m, at rest. The collision is head on and elastic in nature. After the collision the fraction of energy lost by colliding body A is

(1) $\frac{1}{9}$ (2) $\frac{8}{9}$

(3) $\frac{1}{3}$ (4) $\frac{2}{3}$

21. A 5 kg block is released from top of a smooth incline plane as shown in figure.



After compressing the spring by 1 meter, block comes to rest momentarily. Speed of block, just before it touches the spring, is ($g = 10$ m/s^2)

(1) $\sqrt{20}$ m/s (2) 4 m/s

(3) 5 m/s (4) $\sqrt{10}$ m/s

22. A particle is moving along x-axis. The potential energy of particle (mass 1 kg) is given by $U = 2x^2 - 5x$ J. Potential energy of particle, when particle is in equilibrium, is

(1) $\frac{+25}{8}$ J (2) $\frac{-25}{8}$ J

(3) 48 J (4) Zero

23. A ball is thrown towards a smooth floor at an angle of incidence ($\theta = 45°$) with speed u. The coefficient of restitution between floor and ball is e. After collision with floor angle between velocity and normal is (No effect of gravity)

(1) $\sin^{-1}\left(\frac{1}{e}\right)$ (2) $\cos^{-1}\left(\frac{1}{e}\right)$

(3) $\tan^{-1}\left(\frac{1}{e}\right)$ (4) $\tan^{-1}(e)$

24. A neutron moving at a speed v undergoes a head on elastic collision with an atom at rest. Mass number of atom is A. After collision velocity of neutron is

(1) $\left(\frac{1-A}{1+A}\right)v$ (2) $\left(\frac{1+A}{1-A}\right)v$

(3) Av (4) Zero

25. A ball is dropped on a floor from a height. Coefficient of restitution is 0.5. After collision with floor ball will rise to height of

(1) 9 m (2) 5 m

(3) 2 m (4) 2.5 m

26. A particle is given an initial velocity u at lowest most point inside a smooth vertical circular track of radius $R = 1$ m such that it is just able to complete the circle in vertical plane. Acceleration of the particle when particle is at top most point is

(1) g (2) $\frac{g}{2}$

(3) $g\sqrt{10}$ (4) Zero

27. The nuclides $^3_1$H and $^3_2$He are

(1) Isobars

(2) Isotones

(3) Isotopes

(4) Isomers

28. Density of a nucleus

(1) Depends upon mass number

(2) Depends upon radius

(3) Depends upon atomic number

(4) Is approximately constant

29. Energy equivalent of 2 g of substance is

(1) 2 MeV

(2) 931 MeV

(3) 18×10^{13} J

(4) 9×10^{13} J

30. In p-n junction having depletion layer of thickness 10^{-6} m the potential across it is 0.5 V. The electric field is

(1) 2×10^5 V/m

(2) 5×10^5 V/m

(3) 0.5×10^{-6} V/m

(4) 5×10^{10} V/m

31. In intrinsic semiconductor at room temperature number of free electrons n_e and holes n_h are related as

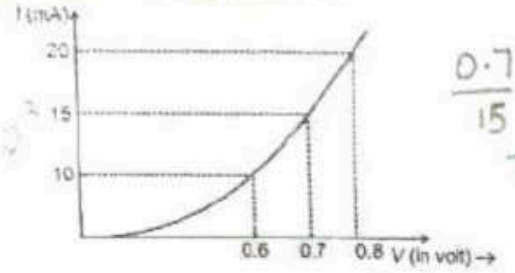
(1) $n_e > n_h$

(2) $n_h > n_e$

(3) $n_h = n_e$

(4) Infinite

32. The V-I characteristic of a silicon diode in forward bias is shown in figure. The dynamic resistance of diode at current $I = 15$ mA is



(1) 35 Ω (2) $\frac{700}{15}$ Ω

(3) 20 Ω (4) 10 Ω

33. If initial activity of a radioactive sample is R_0 and half life of sample is one hour, after 4 hours radioactivity will be

(1) $\frac{R_0}{8}$ (2) $16R_0$

(3) $\frac{R_0}{16}$ (4) $\frac{R_0}{4}$

34. We are given the following atomic masses

$^{238}_{92}U = 238.05079$ u $^{4}_{2}He = 4.00260$ u

$^{234}_{90}Th = 234.04363$ u $^{1}_{1}H = 1.00783$ u

$^{237}_{91}Pa = 237.05121$ u

(All symbols have their usual meaning)

Which among the following is true?

(1) $^{238}_{92}U$ cannot spontaneously emit an alpha particle

(2) $^{238}_{92}U$ can spontaneously emit a proton

(3) $^{238}_{92}U$ cannot spontaneously emit a proton

(4) $^{238}_{92}U$ can emit proton and alpha particles

35. On bombarding U^{235} nucleus by a slow neutron, 200 MeV of energy is released. If the output of atomic reactor is 3.2 MW, then rate of fission will be

(1) 1×10^{17}/s

(2) 2×10^{17}/s

(3) 5×10^{17}/s

(4) 2×10^{15}/s

SECTION - B

36. A bullet of mass 0.05 kg moving with a speed of 100 m/s enters a heavy wooden block and is stopped after a distance of 50 cm. Average acceleration of bullet throughout the journey in block is

(1) -1×10^4 m/s² (2) 1×10^4 m/s²

(3) -2×10^2 m/s² (4) -3×10^2 m/s²

37. A monkey of mass 60 kg climbs on a rope with upward acceleration 5 m/s². Rope is attached to a hook in the ceiling. Tension in the rope is ($g = 10$ m/s²)

(1) 100 N (2) 300 N

(3) 600 N (4) 900 N

38. The rear side of a truck is open and a box of 50 kg mass is placed on the surface of truck. The coefficient of friction between the box and the surface below it is 0.2. The maximum acceleration of truck on horizontal road, so that box does not fall off the truck, is ($g = 10$ m/s²)

(1) 1 m/s² (2) 2 m/s²

(3) 4 m/s² (4) 10 m/s²

39. A particle of mass $3m$ at rest suddenly explodes on its own into three identical fragments. Two fragments of mass m each move along mutually perpendicular direction with speed v each. The energy released during the process is

(1) $2mv^2$ (2) $\frac{1}{2}mv^2$

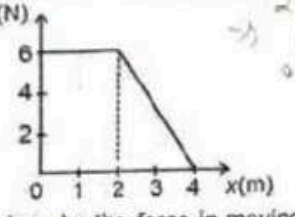
(3) $\frac{3}{2}mv^2$ (4) mv^2

40. A force $F = 3 + 4x$ acts on particle in x direction, where F and x are in SI unit. Work done by this force to move particle from $x = 0$ to $x = 2$ m is

(1) 4 J (2) 14 J

(3) 2 J (4) –2 J

41. A force (F) acting on an object varies with distance (x) is shown in following figure.



The work done by the force in moving the object from $x = 1$ m to $x = 4$ m is

(1) 12 J (2) 18 J

(3) 6 J (4) –6 J

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42. A body of mass 1 kg begins to move under the action of a time-dependent force $\vec{F} = (4t\hat{i} + 6t^2\hat{j})$ N, where $\hat{i}$ and $\hat{j}$ are unit vector along x and y axes. Power delivered by the force at $t = 1$ s is

(1) 5 W
(2) 10 W
(3) 30 W
(4) 20 W

43. Two stable isotopes of nucleus X (^{6}X and ^{7}X) have respectively abundances of 10% and 90%. These isotopes have masses 6.01512 u and 7.01600 u respectively. Atomic mass of X is

(1) 6.91591 u
(2) 6.441 u
(3) 7.00140 u
(4) 6.9315 u

44. The binding energy of deuteron is 2.15 MeV and that of $^{4}_{2}He$ is 28.05 MeV. If two deuterons are fused to form one $^{4}_{2}He$, then the energy released is

(1) 28.75 MeV
(2) 23.75 MeV
(3) 25.75 MeV
(4) 26.15 MeV

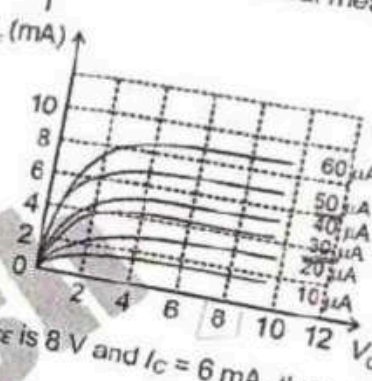
45. Potential energy of a pair of nucleons may be
(A) Positive
(B) Negative
(C) Zero
(1) (A) Only
(2) (A) and (B) only
(3) (B) and (C) only
(4) (A), (B) and (C)

46. Suppose a pure Si crystal has 5×10^{28} atoms/m^3. If it is doped by 2 ppm concentration of Al, then electron density is (Given $n_i = 1.5 \times 10^{16}$ m^{-3})
(All symbols have their usual meaning)
(1) 2.25×10^{9} /m^3
(2) 1×10^{23} /m^3
(3) 5×10^{8} /m^3
(4) 1.5×10^{16} /m^3

47. A p-n photodiode is made of material with a band gap of 1.5 eV. The maximum wavelength of the radiation that can be absorbed by the material is nearly.
(1) 6267 Å
(2) 6200 Å
(3) 8957 Å
(4) 8267 Å

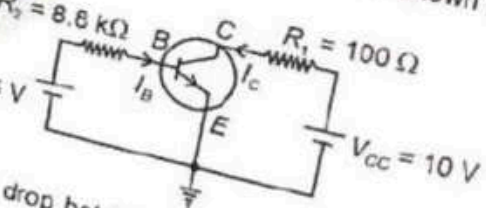
48. In a transistor the collector current is always less than the emitter current because
(1) A few charge carriers are lost in the base as only remaining ones reach the Collector
(2) Length of collector is greater than that of emitter
(3) Collector being reverse biased, attract less electrons
(4) Collector side is forward biased and emitter side is reverse biased

49. The output characteristics of an n-p-n transistor in common emitter configuration is shown in figure. (All symbols have their usual meaning)



When V_{CE} is 8 V and $I_C = 6$ mA, then value of β_{ac} is
(1) 99
(2) 200
(3) 105
(4) 33

50. A silicon transistor amplifier circuit is shown in the figure.



Voltage drop between base and emitter is 0.6 V and $\beta = 100$. Then match column I with column II and choose correct option.
(All units are measured in SI unit system, symbol have their usual meaning)

(A)	Base current I_B	(P)	50×10^{-3}
(B)	Collector current I_C	(Q)	0.5×10^{-3}
(C)	Collector-emitter voltage	(R)	5
(D)	Collector-base voltage	(S)	4.4

(1) (A)-(P), (B)-(Q), (C)-(R), (D)-(S)
(2) (A)-(Q), (B)-(P), (C)-(R), (D)-(S)
(3) (A)-(Q), (B)-(P), (C)-(S), (D)-(R)
(4) (A)-(S), (B)-(P), (C)-(R), (D)-(Q)

CHEMISTRY

SECTION - A

51. Which among the following is a non-reducing sugar?
(1) Glucose (2) Fructose
(3) Ribose (4) Sucrose

52. Both temporary and permanent hardness can be removed by
(1) Boiling
(2) Treatment with Na_2CO_3
(3) Zeolites
(4) Clark's method

53. Which of the following is an acidic amino acid?
(1) Glycine (2) Aspartic acid
(3) Leucine (4) Lysine

54. Deficiency of which vitamin causes scurvy disease?
(1) Vitamin E (2) Vitamin D
(3) Vitamin C (4) Vitamin A

55. Total number of non-essential amino acid among the following is

Glycine, alanine, valine, aspartic acid, serine and tyrosine

(1) 3 (2) 4
(3) 5 (4) 6

56. Consider the following statements given below
(a) Starch is the main storage polysaccharide of plants
(b) Starch is a polymer of β - D - (+) Glucose made up of two components namely amylose and amylopectin
(c) Amylose is water soluble
(d) Amylopectin is a branched chain polysaccharide

Correct statements are
(1) a and b only (2) a, b and c only
(3) a, c and d only (4) a, b and d only

57. Which of the following is not a natural polymer?
(1) Starch (2) Cellulose
(3) Proteins (4) Buna – S

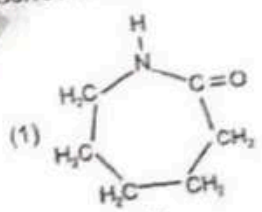
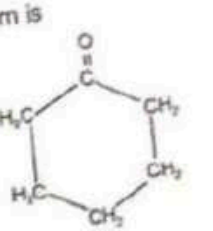
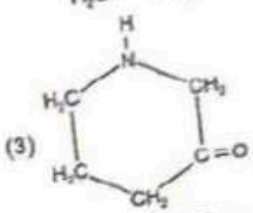
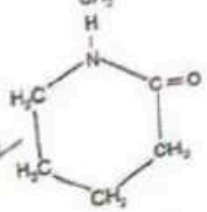
58. Glycine and amino caproic acid are monomer of
(1) Nylon - 6, 6 (2) Nylon 2 - Nylon 6
(3) PHBV (4) Buna - N

59. Which of the following is a fat-soluble vitamin?
(1) Vitamin – B_1 (2) Vitamin – C
(3) Vitamin – D (4) Vitamin –B_2

60. Which one of the following polymers is prepared by addition polymerisation?
(1) Teflon (2) Bakelite
(3) Glyptal (4) Dacron

61. **Statement I** : The low density polythene has highly branched structure.
Statement II : The high density polythene has linear structure
In the light of above statements choose the correct option from the given below options
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

62. Correct structure of caprolactam is

(1) 
(2) 
(3) 
(4) 

63. $\left[CH_2-CH(CN)\right]_n$ is a
(1) Copolymer
(2) Condensation polymer
(3) Homopolymer
(4) Natural polymer

64. Which of the following is a cationic detergent?
(1) Sodium stearate
(2) Cetyltrimethyl ammonium bromide
(3) Sodium lauryl sulphate
(4) Sodium dodecylbenzene sulphonate

65. Which artificial sweetener is trichloro derivative of sucrose?
(1) Aspartame (2) Saccharin
(3) Sucralose (4) Alitame

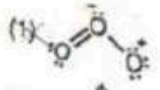
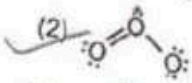
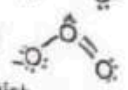
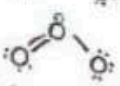
66. Drugs that bind to the receptor site and inhibit its natural function are called
(1) Agonists (2) Antagonists
(3) Enzymes (4) Receptors

67. Antihistamine drug among the following is
(1) Cimetidine (2) Ranitidine
(3) Terfenadine (4) Histamine

68. Which among the following is an example of barbituric acid derivatives?
(1) Veronal (2) Chlordiazepoxide
(3) Meprobamate (4) Equanil

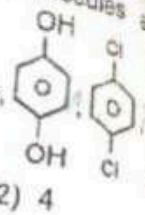
69. Which one of the following is employed as a non-narcotic analgesics?
(1) Morphine (2) Heroin
(3) Codeine (4) Aspirin

70. In CO molecule, the bond order is
(1) 2.5 (2) 3
(3) 2 (4) 3.5

71. The correct Lewis structure of O_3 molecule is
(1) 
(2) 
(3) 
(4) 

72. Which among the following is an odd-electron molecule?
(1) NO_2 (2) BCl_3
(3) CO (4) BeF_2

73. The correct order of covalent bond length is
(1) O – H < C – H < C = O < C ≡ C
(2) O – H < C = O < C – H < C ≡ C
(3) C ≡ C < C = O < C – H < O – H
(4) C = O < C ≡ C < O – H < C – H

74. Total number of polar molecules among the following is HF, H_2O, CO_2, 
(1) 3 (2) 4
(3) 5 (4) 2

75. Identify the incorrect statement among the following regarding chemical properties of hydrogen.
(1) Dihydrogen can reduce chlorine into chloride (Cl^-).
(2) Dihydrogen is reduced by sodium to NaH.
(3) Dihydrogen oxidized copper to copper (II) state.
(4) Dihydrogen can reduce Pd^{2+} to Pd.

76. T-shaped molecule among the following is
(1) SF_4 (2) ClF_3
(3) BrF_5 (4) XeF_4

77. Which physical property of H_2O is higher than that of D_2O?
(1) Melting point
(2) Density (298 K)
(3) Dielectric constant
(4) Enthalpy of vaporization (373 K)

78. Hybridization of Ni in $[Ni(CN)_4]^{2-}$ is
(1) sp^3 (2) dsp^2
(3) sp^3d (4) sp^2

79. Paramagnetic molecule among the following is
(1) Li_2 (2) B_2
(3) N_2 (4) F_2

80. Consider the given reaction,
(i) $I_2 + H_2O_2 + OH^- \longrightarrow A + B + O_2$
Products A and B respectively are
(1) I^-, H_2 (2) IO_3^-, H_2O
(3) IO_3^-, H_2 (4) I^-, H_2O

81. Species having maximum number of pi (π) bonds is
(1) C_2H_6 (2) C_2H_4
(3) O_2 (4) C_2

82. Hydrogen bonding is not possible in

(1) C_2H_5OH (2) NH_3

(3)  (4) BH_3

83. Among the following, which compound has maximum lattice energy?

(1) AlN (2) MgO

(3) KCl (4) NaF

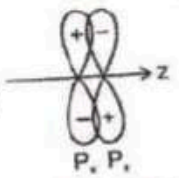
84. Which among the following species contains two $p\pi - d\pi$ bonds?

(1) XeO_3 (2) SO_3

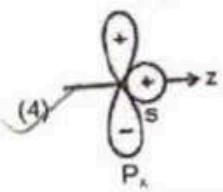
(3) SO_2 (4) CO_2

85. The correct representation for zero overlapping among the following is

(1) 

(2) 

(3) 

(4) 

SECTION - B

86. Consider the following statements regarding hydrogen.

(a) In term of ionization enthalpy, hydrogen resembles more with alkali metals

(b) H^+ does not exist freely

(c) All three isotopes of hydrogen have same rate of reactions.

Incorrect statement(s) is/are

(1) a only (2) a and b only

(3) a and c only (4) c only

87. **Statement I** : Process of producing syngas from coal is called coal gasification.

Statement II : Commercially, H_2 is prepared by the reaction of zinc with aqueous alkali.

In the light of above statements, choose the correct option given below.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

88. Consider the statements regarding VSEPR theory

(a) The shape of a molecule depends upon the number of valence shell electron pairs around the central atom.

(b) The pairs of electrons tend to occupy such positions in space which leads to minimum repulsion between them

(c) The repulsive interaction of lone pair –lone pair is less than lone pair - bond pair repulsion.

Correct statement(s) is/are

(1) a only (2) a and b only

(3) a and c only (4) c only

89. Which of the following is electron-rich hydride?

(1) B_2H_6 (2) CH_4

(3) SiH_4 (4) NH_3

90. Correct order of bond angle is

(1) $CH_4 < NH_3 < N_2O$ (2) $NH_3 < CH_4 < H_2O$

(3) $NH_3 < H_2O < CH_4$ (4) $H_2O < NH_3 < CH_4$

91. **Statement I**: In $CuSO_4 \cdot 5H_2O$, water molecules are associated with hydrogen bonds as well as coordinate bonds.

Statement II: In $CuSO_4.5H_2O$, only one water molecule is associated with hydrogen bonding.

In the light of above statements, choose the correct option given below

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

92. Match the following and identify the correct option.

	Column -I		Column-II
(A)	Saline hydride	(i)	HF
(B)	Electron-precise covalent hydride	(ii)	CrH
(C)	Electron-rich covalent hydride	(iii)	CH_4
(D)	Metallic hydride	(iv)	NaH

(1) (A) - (ii), (B) - (iii), (C) - (i), (D) - (iv)
(2) (A) - (ii), (B) - (i), (C) - (iii), (D) - (iv)
(3) (A) - (iv), (B) - (iii), (C) - (ii), (D) - (i)
(4) (A) - (iv), (B) - (iii), (C) - (i), (D) - (ii)

93. Number of asymmetric carbon in α-D-Fructo furanose is
(1) 3
(2) 4
(3) 5
(4) 6

94. The strength of 20 volume solution of H_2O_2 is approximately
(1) 30 g/L
(2) 60 g/L
(3) 15 g/L
(4) 40 g/L

95. Identify the pair of species having same bond order among the following is
(1) N_2^+, N_2^-
(2) O_2, O_2^+
(3) H_2, H_2^+
(4) O_2^-, O_2^{2-}

96. Identify the incorrect statement regarding rubber is
(1) It is a linear polymer
(2) 2-methyl -1, 3-butadiene is its monomer
(3) It has random cis and trans-configuration of isoprene units
(4) Polymer chains are held together by weak van der Waals interactions

97. The base which is present in RNA but not in DNA is
(1) Adenine
(2) Guanine
(3) Cytosine
(4) Uracil

98. The monomer of PVC polymer is
(1) $CH_2 = \underset{CH_3}{\underset{|}{C}}H$
(2) $CH_2 = \underset{C_6H_5}{\underset{|}{C}}H$
(3) $CH_2 = \underset{Cl}{\underset{|}{C}}H$
(4) $CH_2 = CH_2$

99. The correct increasing order of intermolecular forces of the given polymers is
(1) Nylon 6, 6 < Buna - S < Polythene
(2) Buna – S < Polythene < Nylon 6,6
(3) Polythene < Nylon 6, 6 < Buna - S
(4) Nylon 6, 6 < Polythene < Buna - S

100. Identify the incorrect statement about antiseptics and disinfectants from the following.
(1) Dettol is a mixture of chloroxylenol and terpineol
(2) Iodine is a powerful antiseptic
(3) Boric acid in dilute aqueous solution is weak antiseptic for eyes.
(4) 0.2 percent solution of phenol is disinfectant.

BOTANY

SECTION - A

101. The mycobiont partner in lichen does not
(1) Prepare food for its partner
(2) Provide shelter to its partner
(3) Absorb mineral nutrients for its partner
(4) Absorb water for its partner

102. Select the **incorrect** option for viroids
(1) Cause potato spindle tuber disease
(2) Discovered in 1971
(3) Nucleic acid is protected by capsid
(4) Its RNA was of low molecular weight

103. All of the following are criteria for classification given by R. H. Whittaker, **except**
(1) Mode of nutrition
(2) Type of locomotory organ
(3) Phylogenetic relationship
(4) Cell structure

104. Select the **incorrect** feature for Euglenoids
(1) Presence of two unequal flagella
(2) Majority found in fresh water
(3) Cell wall is present
(4) Have pigments identical to those in higher plants

105. Consider the following characteristics and identify the correct fungi on the basis of them.
A. Branched and septate mycelium
B. Sexual spores are produced exogenously
C. Sex organs are absent
D. Generally asexual spores are absent
(1) *Neurospora* (2) *Ustilago*
(3) *Alternaria* (4) *Rhizopus*

106. Chemosynthetic autotrophic bacteria
(1) Fix atmospheric nitrogen in heterocyst
(2) Are most abundant in nature
(3) Play great role in recycling nutrients like phosphorus.
(4) Cannot synthesize ATP

107. In which fungi, hyphae are continuous tubes, filled with multinucleated cytoplasm?
(1) *Penicillium* (2) *Mucor*
(3) *Aspergillus* (4) *Claviceps*

108. Consider the following statements and choose the correct option.
Statement A: Bladderwort and *Cuscuta* are insectivorous plants.
Statement B: Animals store food reserves as glycogen or fats and show holozoic mode of nutrition.
(1) Only statement A is incorrect
(2) Only statement B is incorrect
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

109. Select the **odd** one out w.r.t. sexual spores
(1) Basidiospores (2) Oospores
(3) Ascospores (4) Sporangiospores

110. Select the **odd** one Protists w.r.t. mode of nutrition.
(1) Dinoflagellates (2) Protozoan
(3) Chrysophytes (4) Euglenoids

111. **Assertion:** In *Ustilago*, dikaryon stage is observed.
Reason: In Basidiomycetes, Karyogamy is delayed after plasmogamy.
(1) Both assertion & reason are true and the reason is the correct explanation of the assertion.
(2) Both assertion & reason are true but the reason is not the correct explanation of the assertion.
(3) Assertion is true statement but reason is false.
(4) Both assertion and reason are false statements.

112. Select the **correct** match
(1) Amoeboid protozoans – Some of them are autotrophs
(2) Flagellated protozoans – Form symbiotic relationship
(3) Ciliated protozoans – Have gullet
(4) Sporozoans – They are non-infectious

113. The common characteristics shared by member of Basidiomycetes and Deuteromycetes is :
(1) Asexual reproduction by conidia
(2) Absence of sex organs
(3) Vegetative reproduction is absent
(4) Majority of the members are parasites

114. Which of the following is not a feature of dinoflagellates?
(1) Can behave like heterotrophs by predating animals
(2) Presence of chlorophyll a
(3) Presence of stiff cellulosic cell plates
(4) Few forms are toxic for the marine lifeforms

115. Read the following statements and select the **correct** ones.

A. In general, plant viruses exhibit the presence of ssDNA.

B. Arrangement of capsomeres defines the shape of virus.

C. Viruses are obligate parasites.

D. Viruses have inert crystalline structure inside the host

(1) A and C only

(2) B and C only

(3) A and D only

(4) B and D only

116. Members of Ascomycetes

(1) Are saprophytic or coprophilous

(2) Produce exogenous sexual spores

(3) Are called club fungi

(4) Have fruiting body called asci

117. Indian Botanical Garden is situated in

(1) Howrah

(2) England

(3) Lucknow

(4) Delhi

118. Which among the following is **not** a universal rule of binomial nomenclature?

(1) Biological names are generally in Latin and written in italics.

(2) First word in the biological name represents the genus while the second component denotes specific epithet.

(3) Both words in a biological name when handwritten are separately underlined.

(4) Both words of a biological name, genus and specific epithet start with capital letter.

119. All of the following are orders, **except**

(1) Primata

(2) Insecta

(3) Diptera

(4) Poales

120. Match the following column I with column II and select the **correct** option.

	Column I		Column II
a.	Herbarium	(i)	Collection of living plants for reference.
b.	Museum	(ii)	A protected environment for wild animals under human care.
c.	Botanical garden	(iii)	Quick referral system in taxonomic studies.
d.	Zoological parks	(iv)	Collection of preserved plants and animals specimen for study.

(1) a(iv), b(iii), c(i), d(ii)

(2) a(iii), b(i), c(iv), d(ii)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iii), b(i), c(ii), d(iv)

121. Taxonomical keys are

(1) Used for identification of plants only based on similarities.

(2) Separate for each taxonomic category

(3) Generally theoretical in nature

(4) Based only on dissimilarities among species

122. The scientific names of plants are based on agreed principle and criteria which are provided in

(1) International Code for Biological Names

(2) International Criteria for Binomial Nomenclature

(3) International Code for Botanical Nomenclature.

(4) International Code for Binomial Nomenclature.

123. Select the **incorrect** match among the following.

(1) *Mangifera* – Genus

(2) Dicotyledonae – Class

(3) Muscidae – Order

(4) Poaceae – Family

124. Which of the following is not a measure to control global warming?
(1) Slowing down growth of human population.
(2) Reducing reforestation
(3) Cutting down use of fossil fuels
(4) Improving efficiency of energy use

125. **Statement I:** Human excreta can be recycled into a resource without use of water which reduces need of chemical fertilisers.
Statement II: It is practical, hygienic, efficient and cost effective solution to human waste disposal.
Above statements are related to all, **except**
(1) Dry composting of human excreta
(2) EcoSan toilets
(3) Sustainable system of handling human excreta.
(4) Sewage Treatment Plant

126. All of the following are the problems that have come in the wake of green revolution, **except**
(1) Waterlogging due to improper drainage of water.
(2) Soil salinity due to waterlogging
(3) Excessive use of chemicals
(4) Desertification

127. Greenhouse gas which has a major contribution towards global warming is
(1) Carbon monoxide
(2) Carbon dioxide
(3) Methane
(4) Nitrous oxide

128. Mark the **correct** statement.
(1) Nuclear energy has been a non-polluting way for generating electricity.
(2) Greenhouse gases absorb long wave radiation from the earth.
(3) Thickness of ozone is measured in decibels.
(4) Diesel is better fuel than CNG.

129. According to Central Pollution Control Board (CPCB), what size (in diameter) of particulate is responsible for causing greatest harm to human health?
(1) 7.5 μm
(2) 25 μm
(3) 15 μm
(4) 2. 5 μm

130. Biomagnification in the aquatic food chain can be reported when water is polluted by all of the following, **except**
(1) Waste from metal extraction industry
(2) Water from petroleum industry
(3) Sewage from homes
(4) Water from paper manufacturing units

131. The best solution for treatment of e-waste is
(1) Burrying in landfills
(2) Recycling
(3) Burning
(4) Incineration

132. Select the **correct** match.

(1) Eutrophication	–	Waste disposal
(2) Snow blindness	–	UV-A radiation
(3) Jhum cultivation	–	Conservation strategy for forests
(4) Montreal Protocol	–	Control emission of ozone depleting substances.

133. Ozone hole
(1) Is a large area of thinned ozone layer
(2) Was marked over Arctic region
(3) Is formed in troposphere
(4) Refers to a region where no ozone is present

134. Consider the below statements and select the **correct** option.
Statement I: The use of incinerators is crucial to disposal of hospital waste
Statement II: Electrostatic precipitator can remove gases like sulphur dioxide using spray of water or lime.
(1) Only statement I is incorrect
(2) Only statement I is correct
(3) Both statements are correct
(4) Both statements are incorrect

135. In India, in which year the Air Act was amended to include noise as an air pollutant?
(1) 1981
(2) 1974
(3) 1987
(4) 1986

SECTION - B

136. Which of the following groups exhibits the most extensive metabolic diversity?
(1) Bacteria (2) Fungi
(3) Protista (4) Plantae

137. The three domain system comprises of
(1) Three Kingdoms (2) Four Kingdoms
(3) Five Kingdoms (4) Six Kingdoms

138. All of the following prokaryotes have cell wall, **except**
(1) Cyanobacteria
(2) Mycoplasma
(3) *Nostoc*
(4) *Methanobacterium*

139. Archaebacteria differ from eubacteria, due to
(1) Lack of nuclei
(2) Cellular level of organization
(3) Different cell wall structure
(4) Heterotrophic mode of nutrition

140. Which of the following characteristic is **not** common between slime moulds and animals?
(1) Spores with cell wall
(2) Absence of cell wall
(3) Heterotrophic mode of nutrition
(4) Being eukaryotic

141. The chief 'producers' in the oceans are
(1) Dinoflagellates (2) Diatoms
(3) Euglenoids (4) Slime moulds

142. 'Sleeping sickness' is caused by:
(1) *Plasmodium* (2) *Paramoecium*
(3) *Entamoeba* (4) *Trypansoma*

143. Select the **correct** order of taxonomic category in ascending order
(1) Genus → Species → Family → Order → Class
(2) Species → Genus → Order → Family → Class
(3) Species → Genus → Family → Order → Class
(4) Species → Genus → Class → order → Family

144. The actual account of habitat and distribution of plants of the given area is found in
(1) Monograph (2) Flora
(3) Catalogue (4) Manuals

145. Which characteristic is exclusively present in the human beings?
(1) Metabolism
(2) Cellular organisation
(3) Reproduction
(4) Self-consciousness

146. In an automobile when an exhaust passes through the catalytic converter then unburnt hydrocarbon and carbon monoxide are changed to
(1) Carbon dioxide and water respectively
(2) Carbon dioxide
(3) Nitrogen gas and carbon dioxide respectively
(4) Nitric oxide and water respectively

147. Petrol is less preferred fuel than CNG because
(1) It is cheaper.
(2) Very little amount of petrol is left unburnt.
(3) It burns less efficiently than CNG.
(4) It is difficult to lay down pipe lines to its distribution than for CNG.

148. Select the **incorrect** feature of the substance manufactured by Ahmad Khan.
(1) When it is mixed with bitumen then it is used to lay roads.
(2) Help to improve life of roads.
(3) It decreases the bitumen's water repellent property.
(4) It is a fine powder of recycled modified plastic.

149. According to National Forest Policy, what percent of forest cover is recommended for plains?
(1) 50% (2) 67%
(3) 1.8% (4) 33%

150. What concept was introduced by Government of India after realising the significance of participation by local communities to protect trees?
(1) Chipko Movement
(2) Jhum Cultivation
(3) Joint Forest Management
(4) Slash and Burn Agriculture

ZOOLOGY

SECTION - A

151. The number of disulphide bonds present between two chains of mature insulin is

(1) Four (2) Three

(3) One (4) Two

152. Genetic modification has

(1) Made crops more sensitive to abiotic stresses

(2) Increased reliance on chemical pesticides

(3) Increased efficiency of mineral usage by plants

(4) Helped to increase post harvest losses

153. Complete the analogy by selecting the **correct** option.

Dipterans : Flies : : Lepidopterans : _____

(1) Armyworm (2) Beetles

(3) *Culex* (4) *Aedes*

154. Choose the **odd** one w.r.t. techniques that enable early detection of disorder.

(1) RDT (2) Serum analysis

(3) PCR (4) ELISA

155. Match column-I with column-II and select the correct option.

	Column-I		Column-II
a.	Gene therapy	(i)	*E. coli*
b.	RNAi	(ii)	Retrovirus
c.	Humulin	(iii)	Transgenic mice
d.	Polio vaccine	(iv)	*Agrobacterium tumefaciens*

(1) a(ii), b(iv), c(i), d(iii)

(2) a(iv), b(ii), c(iii), d(i)

(3) a(iv), b(ii), c(i), d(iii)

(4) a(iii), b(i), c(ii), d(iv)

156. Once an insect ingests the inactive *Bt* toxin, it is converted into an active form of toxin due to

(1) The acidic pH of the gut

(2) The alkaline pH of the gut

(3) Binding to the surface of midgut epithelial cells

(4) The low temperature of the gut

157. Today transgenic models exist for how many of the following human diseases given below in the box?

Rheumatoid arthritis, Cancer, Phenylketonuria, Cystic fibrosis, Alzheimer's, Emphysema

(1) Three (2) Four

(3) Five (4) Six

158. Read the given statements and select the incorrect one.

(1) Insulin can be orally administed to diabetic people.

(2) PCR can detect very low amounts of DNA.

(3) A widely used diagnostic test for AIDS is based on the principle of antigen-antibody interaction.

(4) During chemical safety testing, transgenic animals are made to carry genes which make them more sensitive to toxic substances than non-transgenic animals.

159. A person is suffering from night blindness. Doctor advised him to eat which of the following food?

(1) *Bt* corn (2) Basmati rice

(3) Golden rice (4) Turmeric

160. The proteins encoded by which of the following genes control the cotton bollworms?

(a) *cryIAc* (b) *cryIAb*

(c) *cryIIAb*

Select the **correct** option.

(1) (a) only (2) (b) only

(3) (b) and (c) (4) (a) and (c)

161. Which of the following is incorrect for gene therapy?

(1) It can be used to treat a person born with hereditary disease.

(2) It does gene targeting which involves gene amplification.

(3) If gene isolated from marrow cells producing deleted gene is introduced into cells at early child stage, it could be a permanent cure.

(4) It replaces a defective mutant allele with a functional one.

162. The arrangement of teeth in which they are embedded in a socket of jaw bone is called
(1) Diphyodont (2) Heterodont
(3) Pleurodont (4) Thecodont

163. The oral cavity leads into a structure that serves as a common passage for food and air and is commonly referred to as
(1) Larynx (2) Pharynx
(3) Oesophagus (4) Glottis

164. Which among the following types of teeth possess three roots in humans?
(1) Molars (2) Premolars
(3) Incisors (4) Canines

165. All of the following are modifications of mucosal layer of alimentary canal, **except**
(1) Rugae in the stomach
(2) Villi in the small intestine
(3) Crypts of Lieberkühn
(4) Brunner's glands

166. In human, the largest gland of the body weighs about _____ and situated just below the _____. Select the correct option to fill in the blanks respectively.
(1) 1200 to 1500 gm; abdominal cavity
(2) 1200 to 1500 mg; abdominal cavity
(3) 1200 to 1500 gm; diaphragm
(4) 1200 to 1500 mg; diaphragm

167. Consider the following w.r.t. kwashiorkar and select the incorrect one.
(1) Occurs due to simultaneous deficiency of proteins and calories
(2) Results from replacement of mother's milk by a high calorie-low protein diet in a child more than one year in age
(3) Extensive oedema and swelling is present
(4) Shows wasting of muscles and failure of brain development

168. Which of the following enzymes works at slightly acidic pH?
(1) Salivary amylase
(2) Trypsin
(3) Pepsin
(4) Enterokinase

169. Match the column I with column II and select the correct option.

	Column-I		Column-II
a.	Serosa	(i)	Loose connective tissue containing nerves, blood and lymphs
b.	Muscularis	(ii)	Thin mesothelium with some connective tissues
c.	Sub-mucosa	(iii)	Inner circular and outer longitudinal smooth muscle fibres

(1) a(i), b(ii), c(iii) (2) a(ii), b(iii), c(i)
(3) a(ii), b(i), c(iii) (4) a(i), b(iii), c(ii)

170. Consider the statements A and B and select the correct option.
Statement A: The sub-lingual salivary glands are located inside the buccal cavity.
Statement B: The J-shaped organ that possess gastric glands, is situated in the upper left portion of abdominal cavity.
(1) Both the statements A and B are correct
(2) Both the statements A and B are incorrect
(3) Only B is correct
(4) Only A is correct

171. The respiratory structure of crabs that belongs to the largest phylum of animal kingdom and *Catla* that belongs to class Osteichthyes, respectively, are
(1) Gills and gills
(2) Lungs and gills
(3) Gills and skin
(4) Lungs and lungs

172. Contraction of diaphragm and external intercostal muscles lead to
(1) Decrease in the volume of thoracic cavity
(2) Increase in intrapulmonary pressure more than the atmospheric pressure
(3) Decrease in intrapulmonary pressure in comparison to atmospheric pressure
(4) Forced exhalation of air from the lungs

All of the following are not supported by incomplete cartilaginous rings, **except**

(1) Initial bronchioles

(2) Terminal bronchioles

(3) Alveolar ducts

(4) Alveolar sac

174. Which of the given diseases is characterised by the damage to the alveolar walls and decrease in respiratory surface?

(1) Asthma (2) Emphysema

(3) Asbestosis (4) Silicosis

175. Every 1000 mL of blood that enters the alveoli can deliver 'x' mL of oxygen to the tissues under normal physiological conditions.

Select the correct option for 'x'.

(1) 50 (2) 25

(3) 40 (4) 20

176. The volume of air inspired or expired per minute during a normal respiration under normal physiological condition is about

(1) 1100-1200 mL (2) 4000-4500 mL

(3) 6000-8000 mL (4) 1000-1100 mL

177. The partial pressure of oxygen and carbon dioxide in oxygenated blood and deoxygenated blood, respectively, is

(1) 40 mmHg and 45 mmHg

(2) 95 mmHg and 45 mmHg

(3) 104 mmHg and 40 mmHg

(4) 95 mmHg and 40 mmHg

178. Consider the following statements:

(a) Solubility of CO_2 is 20-25 times higher than that of O_2 in blood.

(b) Terminal bronchioles along with the alveolar ducts constitute the exchange part of respiratory system.

(c) Thoracic chamber is anatomically an air tight chamber.

Select the option with **correct** set.

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

179. Choose the correct sequence of layers forming the diffusion membrane in the lungs.

(1) Thin squamous epithelium of alveoli → Basement substance → Ciliated endothelium of alveolar capillaries

(2) Thin squamous epithelium of alveoli → Basement substance → Endothelium of alveolar capillaries

(3) Basement substance → Cuboidal epithelium of alveoli → Endothelium of alveolar capillaries

(4) Basement substance → Endothelium of alveolar capillaries → Thick columnar epithelium of alveoli

180. The shape of curve obtained when percentage saturation of haemoglobin with oxygen is plotted against the pO_2 is

(1) Parabola (2) Linear

(3) Sigmoid (4) Hyperbola

181. The disease which was seen in Bangladesh during the liberation war and in Ethiopia during the severe drought in mid eighties is

(1) PEM (2) Jaundice

(3) Emphysema (4) Asthma

182. Which among the following is not absorbed by active transport in intestinal lumen?

(1) Glucose (2) Amino acids

(3) Sodium ions (4) Water

183. Trachea is a straight tube extending upto the mid-thoracic cavity which divides at the level of

(1) T_1 (2) T_5

(3) C_7 (4) L_1

184. **Assertion (A):** The main challenge for production of insulin using rDNA techniques was getting insulin assembled into a mature form.

Reason (R): Proinsulin contains an extra stretch called B peptide which is not present in mature form.

In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

185. The pCO_2 in oxygenated blood is
(1) Equal to pCO_2 in alveoli
(2) More than pO_2 in tissues
(3) Equal to pCO_2 in deoxygenated blood
(4) Less than pCO_2 in atmospheric air

SECTION - B

186. Consider the following statements
(a) Respiratory rhythm centre is situated in the pons region of brain.
(b) Pneumotaxic centre can reduce the duration of inspiration.
(c) The role of oxygen in the regulation of respiratory rhythm is quite significant.
Select the option with incorrect statements.
(1) (a) only (2) (c) only
(3) (a) and (c) (4) (b) and (c)

187. CO_2 is carried in association with the most abundant formed element of blood in 'x' percent. Here 'x' is
(1) 97 (2) 70
(3) 20-25 (4) 3

188. All of the following conditions are present in tissues and cause shifting of oxygen dissociation curve to the right, **except**
(1) Low partial pressure of O_2
(2) High partial pressure of CO_2
(3) Low temperature
(4) Decrease in pH

189. The gross calorific value of carbohydrates and the physiologic value of fats, respectively, are
(1) 5.65 kcal/g and 9.0 kcal/g
(2) 4.1 kcal/g and 9.0 kcal/g
(3) 4.0 kcal/g and 9.45 kcal/g
(4) 5.65 kcal/g and 9.45 kcal/g

190. Consider the given steps of final digestion within intestine and select the incorrect one.
(1) Lactose $\xrightarrow{\text{Lactase}}$ Glucose + Galactose
(2) Sucrose $\xrightarrow{\text{Sucrase}}$ Glucose + Glucose
(3) Nucleosides $\xrightarrow{\text{Nucleosidases}}$ Sugars + Bases
(4) Dipeptides $\xrightarrow{\text{Dipeptidases}}$ Amino acids

191. **Assertion (A):** Carboxypeptidase produced by intestinal mucosa acts on trypsinogen.
Reason (R): Trypsinogen is the inactive form of trypsin and requires the activation to perform its function.
In the light of above statements, select the correct option.
(1) (A) and (R) are correct and (R) is the correct explanation of (A).
(2) (A) and (R) is true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

192. Select the **incorrect** statement.
(1) Chief cells secrete the proteolytic enzyme pepsinogen.
(2) HCl is secreted by oxyntic cells present in mucosa of stomach.
(3) About 30% of starch is hydrolysed in the buccal cavity.
(4) Salivary amylase belongs to class I of enzymes.

193. The common bile duct and the pancreatic duct open into the duodenum as 'X' which is guarded by 'Y'.
Here Y, is
(1) Sphincter of Oddi
(2) Pyloric sphincter
(3) Gastro-oesophageal sphincter
(4) Sphincter of Boyden

194. The first transgenic cow produced human protein-enriched milk in the same year when
(1) Eli Lilly prepared genetically engineered insulin.
(2) An American company got patent rights on Basmati rice.
(3) First clinical gene therapy was given to a four-year-old girl.
(4) The first recombinant DNA was constructed.

5. Match the column I with column II and select the correct option.

	Column-I		Column-II
a.	Expiratory reserve volume	(i)	TLC-VC
b.	Expiratory capacity	(ii)	VC-IC
c.	Residual volume	(iii)	IC-TV
d.	Inspiratory reserve volume	(iv)	TV + ERV

(1) a(ii), b(i), c(iv), d(iii)

(2) a(iv), b(ii), c(iii), d(i)

(3) a(ii), b(iv), c(i), d(iii)

(4) a(i), b(iii), c(ii), d(iv)

196. There are an estimated _____ varieties of rice in India alone and _____ documented varieties of Basmati are grown in India.

Choose the option that correctly fills the blanks respectively.

(1) 20,000; 27

(2) 270,000; 20

(3) 200,000; 27

(4) 30,000; 37

197. Choose the incorrect match w.r.t. abbreviations and its full form.

(1)	ADA	–	Adenosine deaminase
(2)	Bt	–	*Bacillus thuringiensis*
(3)	cDNA	–	Complementary DNA
(4)	GEAC	–	Global Engineering Approval Committee

198. Rosie, the transgenic animal is associated with the production of

(1) Human protein enriched milk (24 grams per litre)

(2) Human protein (α-1-antitrypsin) enriched milk

(3) Human alpha-lactalbumin enriched milk

(4) Human protein (interferon-α) enriched milk

199. Over 95 percent of all existing transgenic animals are

(1) Monkey (2) *Musca domestica*

(3) *Macropus* (4) Mice

200. Consider the given statements and select the correct option.

Statement A: The green revolution succeeded in tripling the food supply.

Statement B: Increased yields of crops was mainly due to the use of improved crop varieties during green revolution.

(1) Both statements are correct

(2) Only statement A is correct

(3) Only statement B is correct

(4) Both statements are incorrect

❑ ❑ ❑

Aakash BYJU'S

NBTS CODE-A

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MM : 720

NCERT Booster Program

Time : 3 hrs. 20 min

(*for* NEET-2023)

NBTS - 02

Answers

1. (4)	41. (1)	81. (4)	121. (2)	161. (3)
2. (4)	42. (4)	82. (4)	122. (3)	162. (4)
3. (2)	43. (1)	83. (1)	123. (3)	163. (2)
4. (2)	44. (2)	84. (2)	124. (2)	164. (1)
5. (1)	45. (4)	85. (4)	125. (4)	165. (4)
6. (3)	46. (1)	86. (3)	126. (4)	166. (3)
7. (4)	47. (4)	87. (2)	127. (2)	167. (1)
8. (3)	48. (1)	88. (2)	128. (2)	168. (1)
9. (2)	49. (2)	89. (4)	129. (4)	169. (2)
10. (1)	50. (2)	90. (4)	130. (3)	170. (3)
11. (4)	51. (4)	91. (1)	131. (2)	171. (1)
12. (1)	52. (3)	92. (4)	132. (4)	172. (3)
13. (3)	53. (2)	93. (2)	133. (1)	173. (1)
14. (4)	54. (3)	94. (2)	134. (2)	174. (2)
15. (2)	55. (3)	95. (1)	135. (3)	175. (1)
16. (2)	56. (3)	96. (3)	136. (1)	176. (3)
17. (1)	57. (4)	97. (4)	137. (4)	177. (2)
18. (3)	58. (2)	98. (3)	138. (2)	178. (3)
19. (3)	59. (3)	99. (2)	139. (3)	179. (2)
20. (2)	60. (1)	100. (4)	140. (1)	180. (3)
21. (1)	61. (1)	101. (1)	141. (2)	181. (1)
22. (2)	62. (1)	102. (3)	142. (4)	182. (4)
23. (3)	63. (3)	103. (2)	143. (3)	183. (2)
24. (1)	64. (2)	104. (3)	144. (2)	184. (3)
25. (4)	65. (3)	105. (2)	145. (4)	185. (1)
26. (1)	66. (2)	106. (3)	146. (2)	186. (3)
27. (1)	67. (3)	107. (2)	147. (3)	187. (3)
28. (4)	68. (1)	108. (1)	148. (3)	188. (3)
29. (3)	69. (4)	109. (4)	149. (4)	189. (2)
30. (2)	70. (2)	110. (2)	150. (3)	190. (2)
31. (3)	71. (3)	111. (1)	151. (4)	191. (4)
32. (3)	72. (1)	112. (3)	152. (3)	192. (4)
33. (3)	73. (1)	113. (2)	153. (1)	193. (1)
34. (3)	74. (1)	114. (1)	154. (2)	194. (2)
35. (1)	75. (3)	115. (2)	155. (1)	195. (3)
36. (1)	76. (2)	116. (1)	156. (2)	196. (3)
37. (4)	77. (3)	117. (1)	157. (4)	197. (4)
38. (2)	78. (2)	118. (4)	158. (1)	198. (3)
39. (1)	79. (2)	119. (2)	159. (3)	199. (4)
40. (2)	80. (4)	120. (3)	160. (4)	200. (2)